

BÁO CÁO THỰC HÀNH

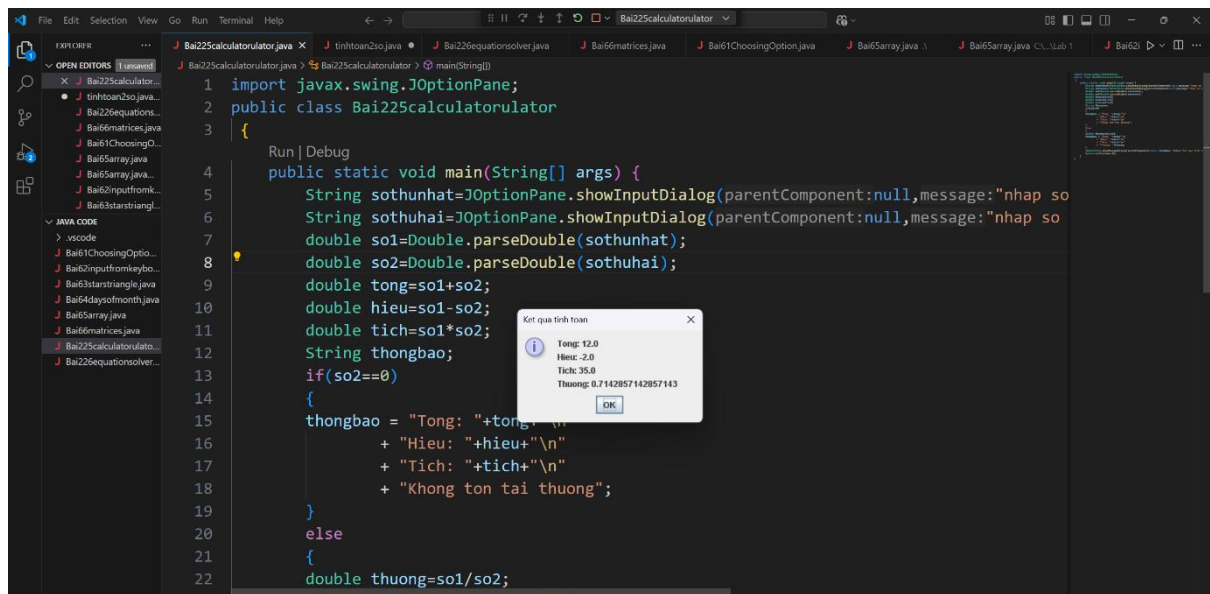
IT3103 – 156772 – LẬP TRÌNH HƯỚNG ĐỐI TƯỢNG

LAB 1

Họ và tên	Đỗ Phú Hưng
Mã số sinh viên	20235738

Bài 2.2.5

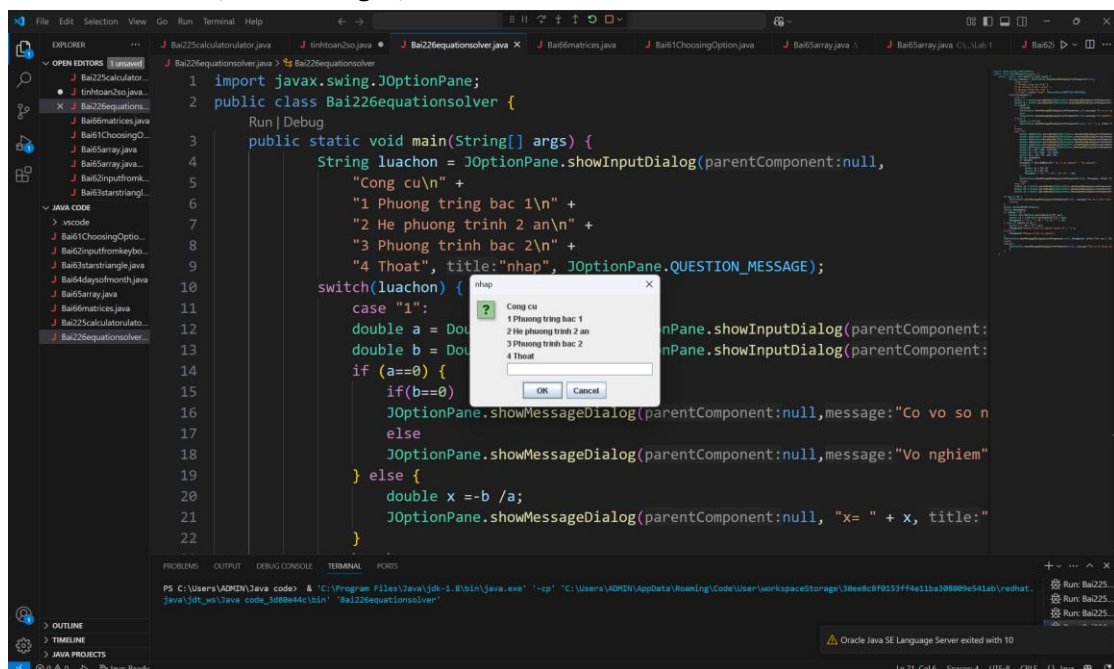
Kết quả sau khi nhập số thứ nhất là 5 và số thứ 2 là 7:



```
1 import javax.swing.JOptionPane;
2 public class Bai225calculatorulator
3 {
4     public static void main(String[] args) {
5         String sothunhat=JOptionPane.showInputDialog(parentComponent:null,message:"nhap so
6         String sothuhai=JOptionPane.showInputDialog(parentComponent:null,message:"nhap so
7         double so1=Double.parseDouble(sothunhat);
8         double so2=Double.parseDouble(sothuhai);
9         double tong=so1+so2;
10        double hieu=so1-so2;
11        double tich=so1*so2;
12        String thongbao;
13        if(so2==0)
14        {
15            thongbao = "Tong: "+tong+"\n"
16            + "Hieu: "+hieu+"\n"
17            + "Tich: "+tich+"\n"
18            + "Khong ton tai thuong";
19        }
20        else
21        {
22            double thuong=so1/so2;
```

Bài 2.2.6

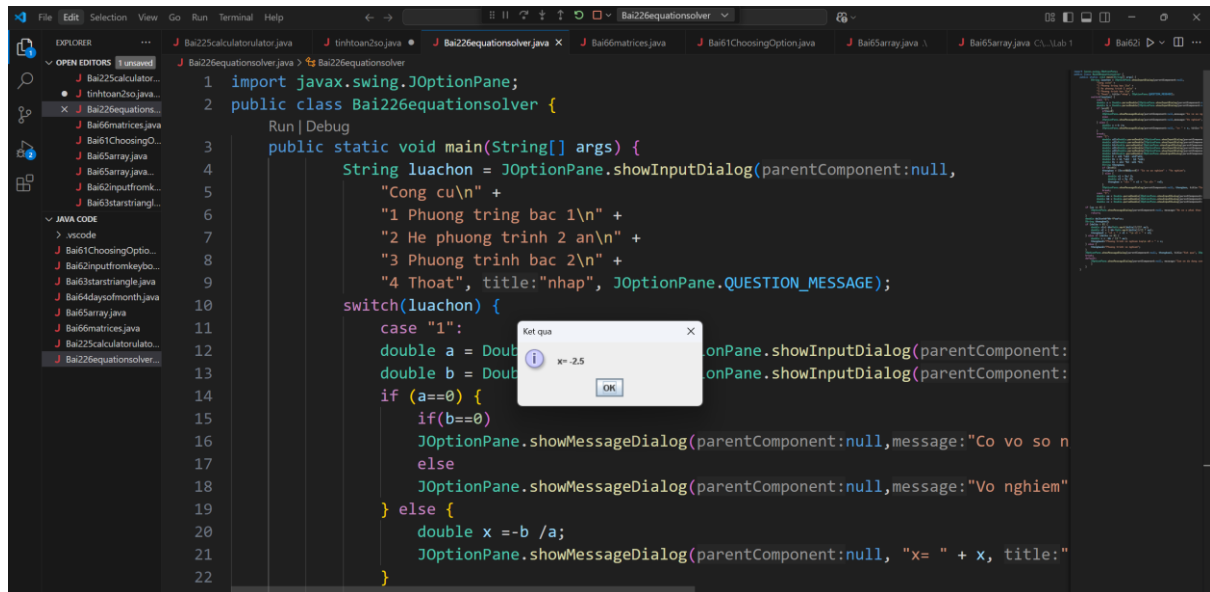
Màn hình hiển thị menu công cụ:



```
1 import javax.swing.JOptionPane;
2 public class Bai226equationsolver {
3     public static void main(String[] args) {
4         String luachon = JOptionPane.showInputDialog(parentComponent:null,
5         "Cong cu\n" +
6         "1 Phuong tring bac 1\n" +
7         "2 He phuong tring 2 an\n" +
8         "3 Phuong tring bac 2\n" +
9         "4 Thoat", title:"nhap", JOptionPane.QUESTION_MESSAGE);
10        switch(luachon) {
11            case "1":
12                double a = Double.parseDouble(JOptionPane.showInputDialog(parentComponent:
13                double b = Double.parseDouble(JOptionPane.showInputDialog(parentComponent:
14                if (a==0) {
15                    if(b==0)
16                    JOptionPane.showMessageDialog(parentComponent:null,message:"Co vo so n
17                    else
18                    JOptionPane.showMessageDialog(parentComponent:null,message:"Vo nghiem"
19                } else {
20                    double x =-b /a;
21                    JOptionPane.showMessageDialog(parentComponent:null, "x= " + x, title:"
22                }
```

Kết quả sau khi chọn công cụ 1 (giải phương trình bậc 1)

+ Chọn a=2, b=5:



```
1 import javax.swing.JOptionPane;
2 public class Bai226equationsolver {
3     public static void main(String[] args) {
4         String luachon = JOptionPane.showInputDialog(parentComponent:null,
5             "Cong cu\n" +
6             "1 Phuong tring bac 1\n" +
7             "2 He phuong tring 2 an\n" +
8             "3 Phuong tring bac 2\n" +
9             "4 Thoat", title:"nhap", JOptionPane.QUESTION_MESSAGE);
10        switch(luachon) {
11            case "1":
12                double a = Double.parseDouble(JOptionPane.showInputDialog(parentComponent:
13                double b = Double.parseDouble(JOptionPane.showInputDialog(parentComponent:
14                if (a==0) {
15                    if(b==0)
16                        JOptionPane.showMessageDialog(parentComponent:null,message:"Co vo so n
17                    else
18                        JOptionPane.showMessageDialog(parentComponent:null,message:"Vo nghiem"
19                } else {
20                    double x = -b / a;
21                    JOptionPane.showMessageDialog(parentComponent:null, "x= " + x, title:"
22                }
23            }
24        }
```

Kết quả sau khi chọn công cụ số 2 (giải hệ phương trình)

a11=1

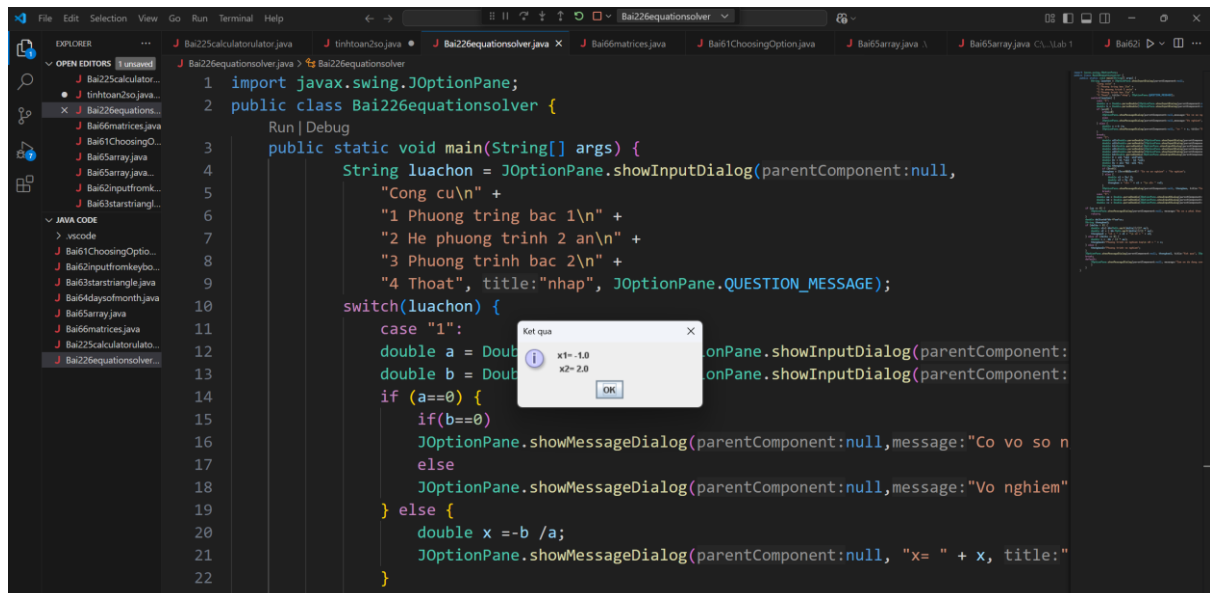
a12=2

b1=3

a21=4

a22=5

b2=6



```
1 import javax.swing.JOptionPane;
2 public class Bai226equationsolver {
3     public static void main(String[] args) {
4         String luachon = JOptionPane.showInputDialog(parentComponent:null,
5             "Cong cu\n" +
6             "1 Phuong tring bac 1\n" +
7             "2 He phuong tring 2 an\n" +
8             "3 Phuong tring bac 2\n" +
9             "4 Thoat", title:"nhap", JOptionPane.QUESTION_MESSAGE);
10        switch(luachon) {
11            case "1":
12                double a = Double.parseDouble(JOptionPane.showInputDialog(parentComponent:
13                double b = Double.parseDouble(JOptionPane.showInputDialog(parentComponent:
14                if (a==0) {
15                    if(b==0)
16                        JOptionPane.showMessageDialog(parentComponent:null,message:"Co vo so n
17                    else
18                        JOptionPane.showMessageDialog(parentComponent:null,message:"Vo nghiem"
19                } else {
20                    double x = -b / a;
21                    JOptionPane.showMessageDialog(parentComponent:null, "x= " + x, title:"
22                }
23            }
24        }
```

Kết quả sau khi chọn công cụ thứ 3

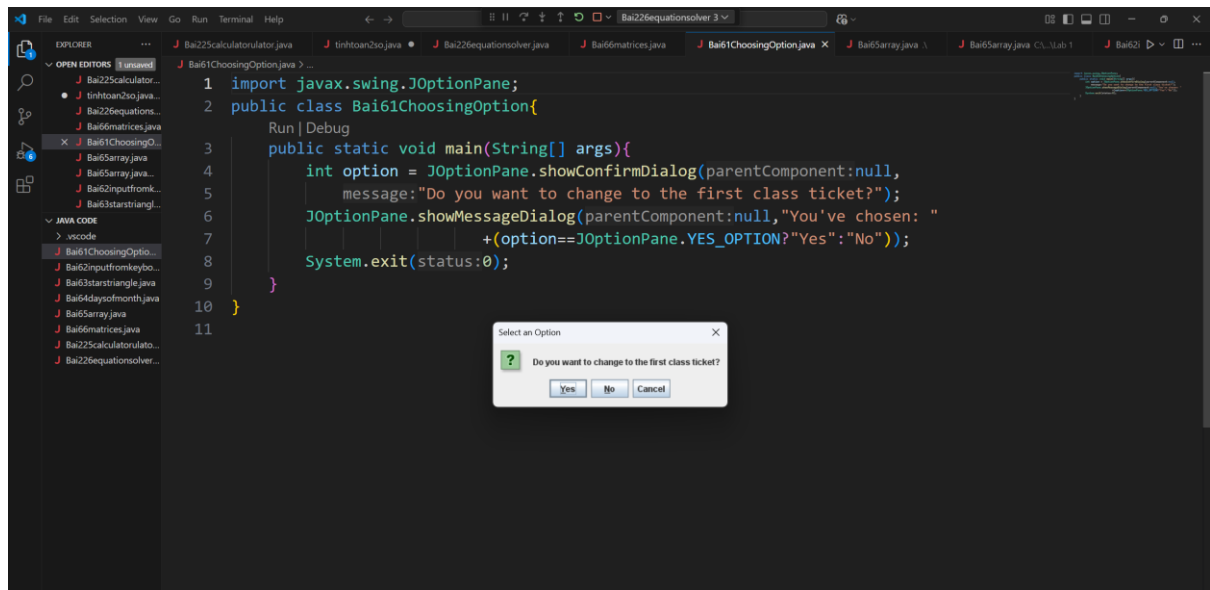
+ Chọn a=1, b=2, c=1

```
2 public class Bai226equationsolver {
3     public static void main(String[] args) {
24         case "2":
25             double a11=Double.parseDouble(JOptionPane.showInputDialog(parentCompon
26             double a12=Double.parseDouble(JOptionPane.showInputDialog(parentCompon
27             double b1=Double.parseDouble(JOptionPane.showInputDialog(parentCompon
28             double a21=Double.parseDouble(JOptionPane.showInputDialog(parentCompon
29             double a22=Double.parseDouble(JOptionPane.showInputDialog(parentCompon
30             double b2=Double.parseDouble(JOptionPane.showInputDialog(parentCompon
31             double D = a11 *a22 -a12*a21;
32             double Dx = b1 *a22 - b2 *a12;
33             double Dy =
34             String tho
35             if (D==0){
36                 thongbao =
37             } else {
38                 double
39                 double
40                 thongt
41                 o parentComponent determines the Frame in which the dialog is displayed; if null ,
42                 or if the parentComponent has no Frame , a default Frame is used
43             JOptionPane.showMessageDialog(parentComponent:null, thongbao, title:"K
44             break;
45         case "3":
```

Kết quả khi chọn lựa chọn 4 (thoát khỏi chương trình)

```
2 public class Bai226equationsolver {
3     public static void main(String[] args) {
24         case "2":
25             double a11=Double.parseDouble(JOptionPane.showInputDialog(parentCompon
26             double a12=Double.parseDouble(JOptionPane.showInputDialog(parentCompon
27             double b1=Double.parseDouble(JOptionPane.showInputDialog(parentCompon
28             double a21=Double.parseDouble(JOptionPane.showInputDialog(parentCompon
29             double a22=Double.parseDouble(JOptionPane.showInputDialog(parentCompon
30             double b2=Double.parseDouble(JOptionPane.showInputDialog(parentCompon
31             double D = a11 *a22 -a12*a21;
32             double Dx = b1 *a22 - b2 *a12;
33             double Dy =
34             String thor
35             if (D==0){
36                 thongbao =
37             } else {
38                 double x1 = Dx/ D;
39                 double x2 = Dy /D;
40                 thongbao = "x1= " + x1 + "\n x2= " +x2;
41             }
42             JOptionPane.showMessageDialog(parentComponent:null, thongbao, title:"K
43             break;
44         case "3":
```

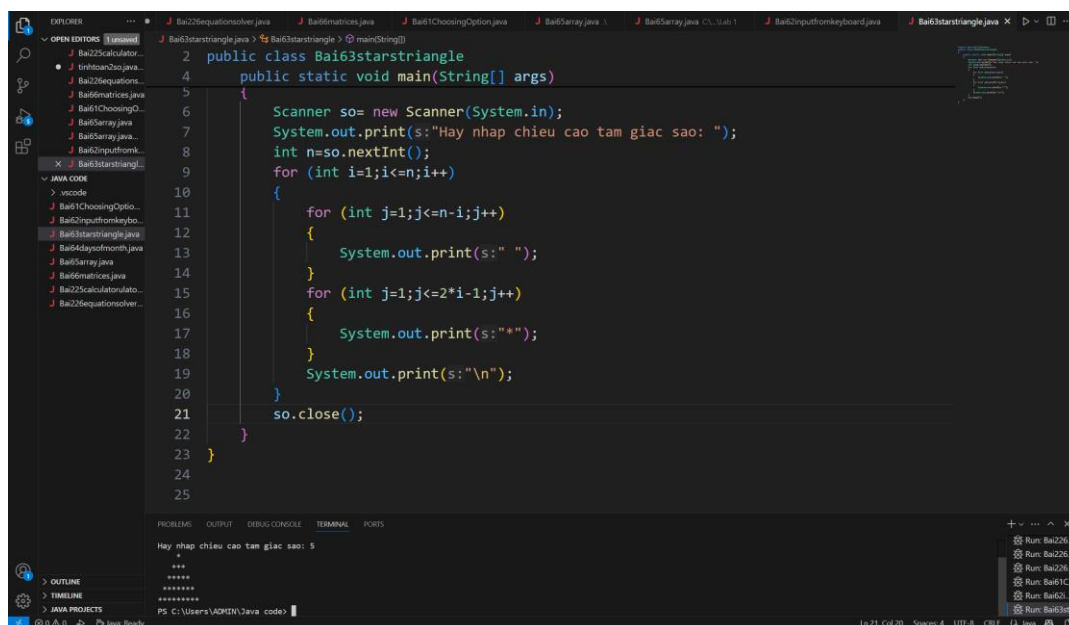
Bài 6.1



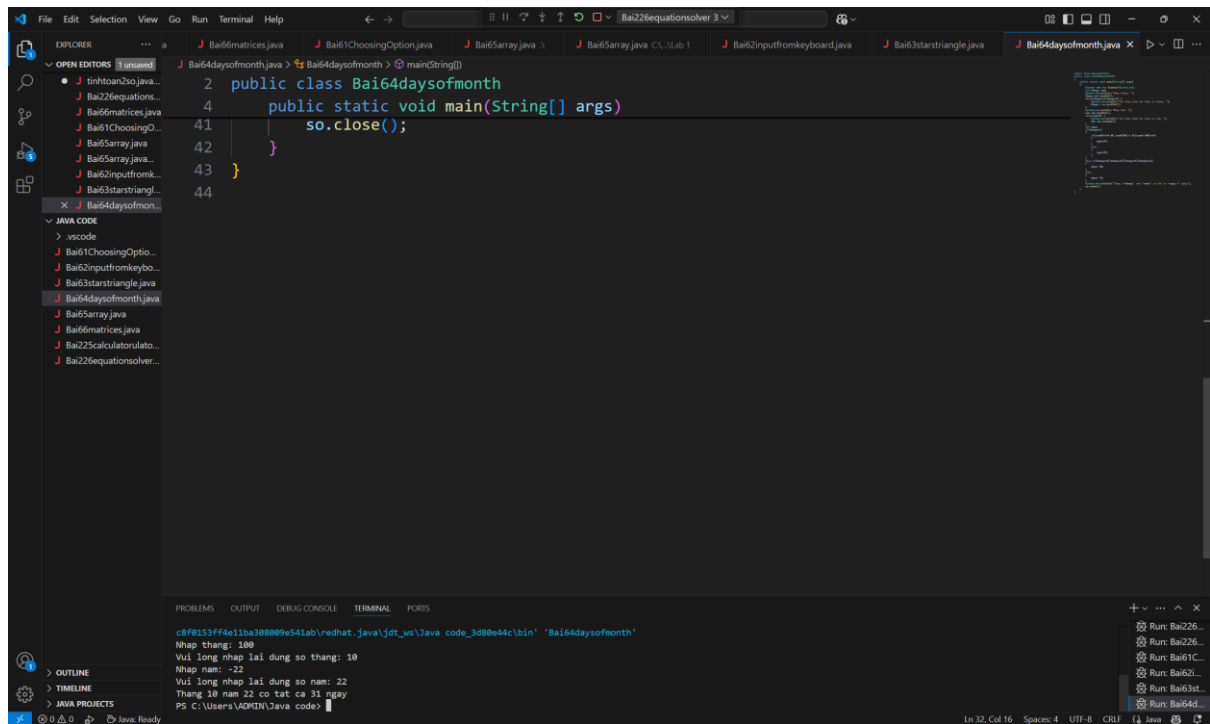
Bài 6.2



Bài 6.3



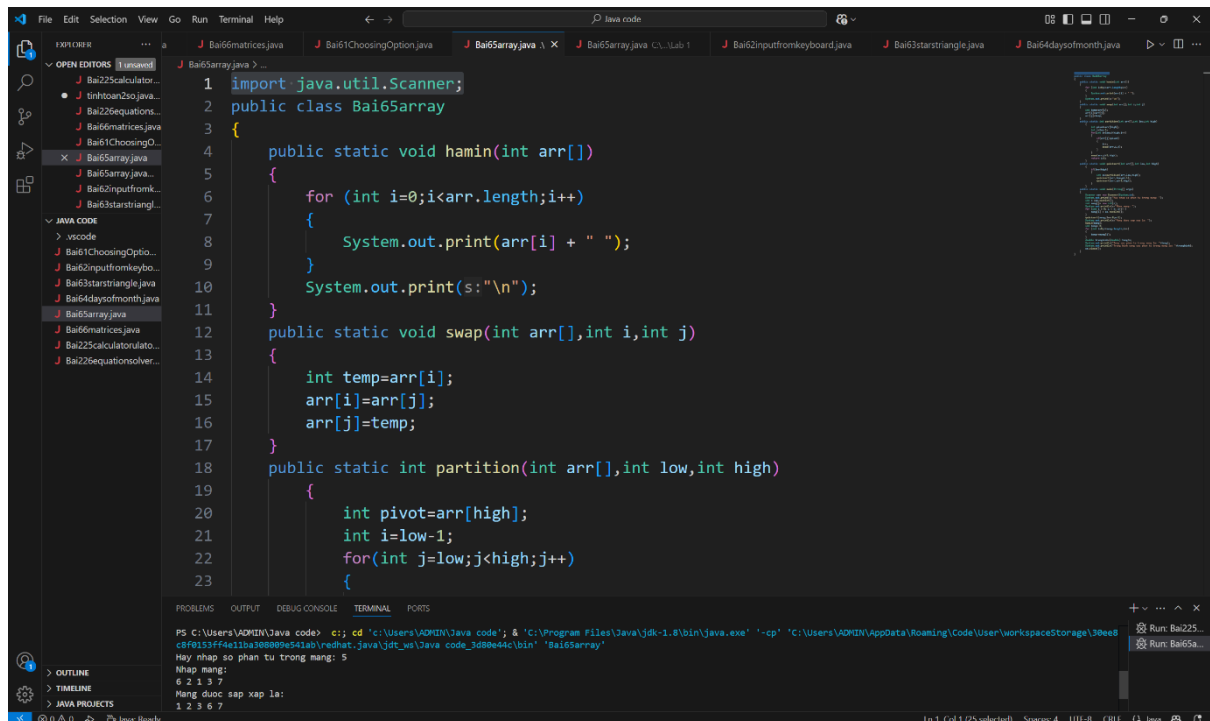
Bài 6.4



```
File Edit Selection View Go Run Terminal Help
Bai226equationsolver 3
EXPLORER
  OPEN EDITORS
    Bai64daysofmonth.java
    Bai66matrices.java
    Bai61ChoosingOption.java
    Bai65array.java
    Bai62inputfromkeyboard.java
    Bai63startriangle.java
    Bai64daysofmonth.java
    Bai65array.java
    Bai66matrices.java
    Bai62calculator.java
    Bai226equationsolver.java
  JAVA CODE
    vscode
    Bai61ChoosingOption.java
    Bai62inputfromkeyboard.java
    Bai63startriangle.java
    Bai64daysofmonth.java
    Bai65array.java
    Bai66matrices.java
    Bai62calculator.java
    Bai226equationsolver.java
  OUTLINE
  TIMELINE
  JAVA PROJECTS
  PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
    c8f0153ff4e11ba308009e541ab\redhat.java\jdk_ws\Java code_3d80e44c\bin\ 'Bai64daysofmonth'
    Nhập tháng: 100
    Vui lòng nhập lại dung so tháng: 10
    Nhập nam: -22
    Vui lòng nhập lại dung so nam: 22
    Tháng 10 nam 22 có tat ca 31 ngay
    PS C:\Users\ADMIN\Java code>

Run: Bai226...
Run: Bai226...
Run: Bai61C...
Run: Bai62...
Run: Bai63st...
Run: Bai64d...
```

Bài 6.5



```
File Edit Selection View Go Run Terminal Help
Java code
EXPLORER
  OPEN EDITORS
    Bai225calculator.java
    Bai226equationsolver.java
    Bai66matrices.java
    Bai61ChoosingOption.java
    Bai65array.java
    Bai62inputfromkeyboard.java
    Bai63startriangle.java
    Bai64daysofmonth.java
    Bai65array.java
    Bai66matrices.java
    Bai62calculator.java
    Bai226equationsolver.java
  JAVA CODE
    vscode
    Bai61ChoosingOption.java
    Bai62inputfromkeyboard.java
    Bai63startriangle.java
    Bai64daysofmonth.java
    Bai65array.java
    Bai66matrices.java
    Bai62calculator.java
    Bai226equationsolver.java
  OUTLINE
  TIMELINE
  JAVA PROJECTS
  PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
    PS C:\Users\ADMIN\Java code> cd 'c:\Users\ADMIN\Java code'; & 'C:\Program Files\Java\jdk-1.8\bin\java.exe' '-cp' 'C:\Users\ADMIN\AppData\Roaming\Code\User\workspaceStorage\30e8...
    c8f0153ff4e11ba308009e541ab\redhat.java\jdk_ws\Java code_3d80e44c\bin\ 'Bai65array'
    Hay nhap so phan tu trong mang: 5
    Nhập mảng:
    6 2 1 3 7
    Mảng được sắp xap là:
    1 2 3 6 7

Run: Bai225...
Run: Bai225...
```

```

Nhap mang:
6 2 1 3 7
Mang duoc sap xap la:
1 2 3 6 7
Tong cac phan tu trong mang la: 19
Trung binh cong cac phan tu trong mang la: 3.8
PS C:\Users\ADMIN\Java code>

```

Bài 6.6

The screenshot shows an IDE with a Java file named `Bai6matrices.java`. The code is as follows:

```

1 import java.util.Scanner;
2 public class Bai6matrices
3 {
4     Run | Debug
5     public static void main(String[] args)
6     {
7         Scanner so= new Scanner(System.in);
8         System.out.println(x:"Nhap so hang cua 2 ma tran: ");
9         int hang=so.nextInt();
10        System.out.println(x:"Nhap so cot cua 2 ma tran: ");
11        int cot=so.nextInt();
12        int a[][]=new int[hang][cot];
13        int b[][]=new int[hang][cot];
14        int tong[][]=new int[hang][cot];
15        System.out.println(x:"Nhap cac phan tu ma tran thu nhât: ");
16        for(int i=0;i<hang;i++)
17        {
18            for(int j=0;j<cot;j++)
19            {
20                a[i][j]=so.nextInt();
21            }
22        }
23        System.out.println(x:"Nhap cac phan tu ma tran thu hai: ");

```

The terminal output shows the execution of the program:

```

PS C:\Users\ADMIN\Java code> & 'C:\Program Files\Java\jdk-1.8\bin\java.exe' "-cp" 'C:\Users\ADMIN\AppData\Roaming\Code\User\workspaceStorage\30ee8c8f0153ff4e11ba388009e541ab\redhat...'
java\jdk\bin\java -cp 'C:\Users\ADMIN\AppData\Roaming\Code\User\workspaceStorage\30ee8c8f0153ff4e11ba388009e541ab\redhat...' Bai6matrices
Nhap so hang cua 2 ma tran:
2
Nhap so cot cua 2 ma tran:
2
Nhap cac phan tu ma tran thu nhât:

```

```

1 1 1 1
Nhap cac phan tu ma tran thu hai:
2 2 2 2
Ma tran la tong cua 2 ma tran tren la:
3 3
3 3

```